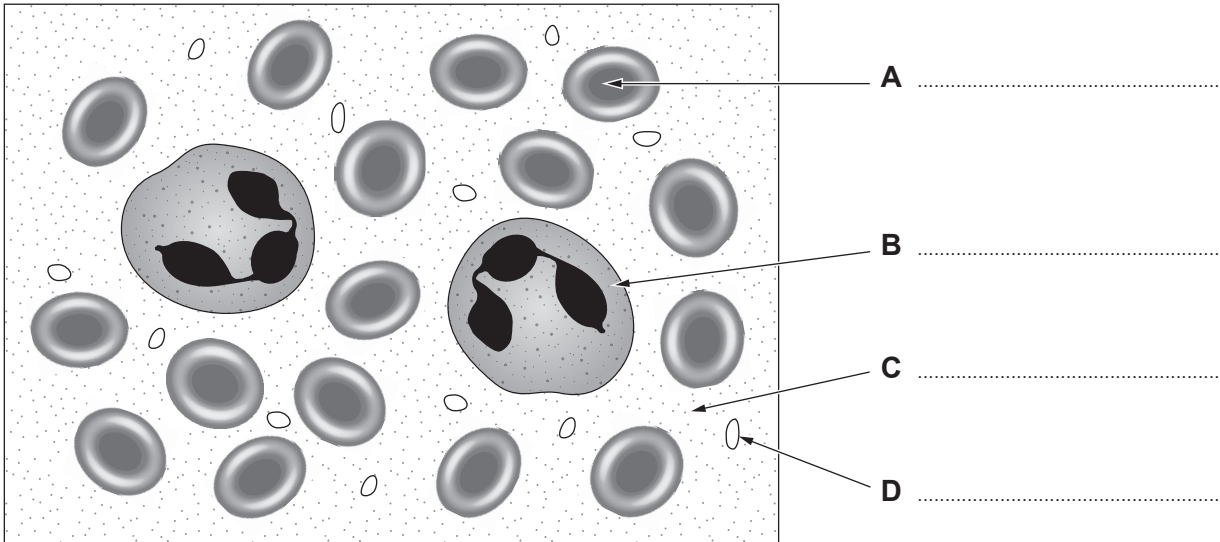


Examiner  
only

2. (a) Blood consists of plasma, red blood cells, platelets and white blood cells. Each part has a different function.

(i) Label the diagram of the blood below. [3]



(ii) Complete the table below. [3]

| Part of the blood | Function                                         |
|-------------------|--------------------------------------------------|
| plasma            | transports carbon dioxide, soluble food and urea |
| red blood cells   | .....                                            |
| platelets         | .....                                            |
| white blood cells | .....                                            |



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(b) Gareth has blood samples taken before a cycle race. His results are shown in the table below together with normal ranges.

|              | Red blood cell count<br>(million / cm <sup>3</sup> ) | White blood cell count<br>(thousand / cm <sup>3</sup> ) | Platelets<br>(thousand / cm <sup>3</sup> ) |
|--------------|------------------------------------------------------|---------------------------------------------------------|--------------------------------------------|
| Normal range | 4.4 - 5.8                                            | 3.9 - 10.8                                              | 130 - 400                                  |
| Gareth       | 2.2                                                  | 6.6                                                     | 105                                        |

(i) Describe how Gareth’s blood sample results compare with the normal range. [2]

.....

.....

.....

(ii) State **one** health problem shown by Gareth’s results. [1]

.....

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(c) Gareth completed a training ride around a cycle track.

- (i) Gareth accelerates from rest to a velocity of 15 m/s in 2.5 seconds.  
Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate his acceleration.

[2]

$$\text{Acceleration} = \dots\dots\dots \text{m/s}^2$$

- (ii) Later in the ride he travels 192 m in 12 s at a constant speed.  
Use the equation:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

to calculate his speed.

[2]

$$\text{Speed} = \dots\dots\dots \text{m/s}$$

- (iii) Gareth cycles 5 laps at this constant speed. Each lap is 500 m.

I. Calculate the distance he travels.

[1]

$$\text{Distance} = \dots\dots\dots \text{m}$$

II. Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the time taken to travel this distance.

[2]

$$\text{Time} = \dots\dots\dots \text{s}$$